UM COLLEGE OF EDUCATION & HUMAN DEVELOPMENT

CHILDREN'S HOSPITALS AND CLINICS OF MINNESOTA (CHC)

Human Factors Research, School of Kinesiology

CHC Minneapolis Campus Infant Care Center, Neonatal Intensive Care Unit, Pediatric Intensive Care Unit, Med/Surg Unit, & Cardiovascular Care Center

Occupancy and Patient Care Quality Benefits of Private Room Designs for Five Different Children's Hospital Intensive Care Units – A Human Factors Evaluation

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How Are Staff Perceptions of Occupancy and Patient Care Quality Affected When Five Different Infant /Child Care Units Are Changed from Multi-Bed (MB) to Private Room (PR) Designs?

Key Conclusions: For the new PR designs: (1) Higher levels of interaction with patients are observed, relative to MB designs; (2) Staff rankings of most attributes of PR occupancy and patient care quality are about the same, or lower than, those for the MB designs; (3) These findings are aligned with staff comments targeting numerous quality defects with the PR patient care environments, plus health problems related to work on these environments; and (4) Collectively, the results suggest that the design, operation and management of the PR patient care environments on the five different units confront a challenge in realizing occupancy and patient care quality benefits that match experience with Neonatal Intensive Care Unit PR designs in other hospitals.

Introduction

►The MB layout represents the most prevalent type of Intensive Care Unit (ICU) design in the U.S., for both children and adult intensive care

►Limited experience with PR Neonatal Intensive Care Units (NICUs) points to both advantages and disadvantages associated with PR relative MB designs

►Advantages:

- more privacy / less noise, disruptions and distraction for patients, parents and staff
- improved patient development and care
- reduced job stress for staff

►Disadvantages:

-reduced interaction among members of patient care team

-lack of global perspective regarding overall status of unit

►Further research is needed to investigate these effects on other types of infant/child ICUs

Background

► In 2010, the CHC Minneapolis campus converted five different infant/child care units to PR designs: an Infant Care Center (ICC), an NICU, a Pediatric Intensive Care Unit (PICU), a Medical/Surgical (Med/Surg) unit, and a Cardio-vascular Care Center (CVCC).

► The NICU provides care for critically ill newborns, emphasizing care for premature infants. The ICC specializes in care of fragile, longer-term neonates and infants up to one year of age. The PICU provides care for critically-ill infants, children, and adolescents, up to age 18.

► The Med/Surg unit provides medical/post-surgical care for pediatric patients in less critical condition, again up to age 18. The CVCC provides pediatric cardiovascular care for fetuses, newborns, infants, children, and adolescents, as well as for adult patients with pediatric heart conditions.

Purpose of Study

To compare and contrast the influence of PR designs versus MB designs on staff perceptions of occupancy and patient care quality on the five different units

Premise and Rationale

►There are interdependent relationships between patient care and patient safety on infant/child ICUs, staff performance in these environments, and staff perceptions of occupancy and patient care quality in these environments --- improvements in the latter should benefit the former

►This is the first study of occupancy and patient care quality for PR versus MB designs for ICC, PICU, Med/Surg and CVCC patient care environments

Views of MB versus PR Patient Care Rooms

MB ICC Room



PR ICC Room



MB NICU Room



PR NICU Room



MB PICU Room



PR PICU Room



Study Design

►This study assessed nursing staff perceptions of occupancy and patient care quality before and after five different infant/child care units on the CHC Minneapolis campus were converted to PR patient care environments

► Independent measures:

-- Pre–post design. Data collected before and after the five units were converted from MB to PR designs

-- Type of unit

►Dependent measures:

-- Task activity analysis observations

-- Staff perceptions of different attributes of occupancy and patient care quality, assessed by an occupancy and patient care quality perceptual response survey questionnaire

-- Staff comments on strengths vs short-comings of PR vs MB patient care room designs

►Null hypothesis:

No effect of MB versus PR patient care room designs, and of type of unit, on dependent measures, for staff employed on the five different units

Methods

Task Activity Analysis Observations

► Fourteen different patient care task activities performed by staff on the five different units were recorded by visual observation

► Across all five units, task activities of a total of 128 nursing staff, all female, were observed during about 123 hours of observation. Numbers of staff observed on each unit were approximately equal.

► Based on these observations, relative frequencies and rates (per minute) of each observed activity were calculated

Perceptual Response Survey

► 127-item instrument

► All survey respondents were female

► MB phase response rates: 7% (ICC respondents) to 15% (Med/Surg respondents)

► PR phase response rates: 18% (CVCC respondents) to 39% (Med/Surg respondents)

► Respondents asked to provide quality rankings (7-level Likert scale) in 13 major areas, termed <u>major indicators</u> of occupancy & patient care quality:

(1) overall employment quality;
(2) level of job satisfaction;
(3) level of work environment comfort;
(4) level of patient care effectiveness;
(5) physical work environment quality;
(6) patient care quality;
(7) job quality;
(8) patient safety quality;
(9) staff safety quality;
(10) quality of interaction among patient care team members;
(11) quality of interaction with patient care technology;
(12) quality of life off-the-job; and
(13) overall quality of personal health.

Staff Comments

► During the PR phase, staff were asked to comment on strengths and shortcomings of PR designs relative to MB room designs

► Across the five units, during the PR phase:

- 774 distinctly different comments (and 1154 total comments) recorded from a total of 206 different staff (comprising over one-third of the 551 staff employed across the five units)
- About 45 percent were negative comments, identifying occupancy and patient care quality concerns or shortcomings of the PR patient care environments

Results

Fig 1. Comparison of PR versus MB Results for Observed Task Activity Frequencies on the ICC.

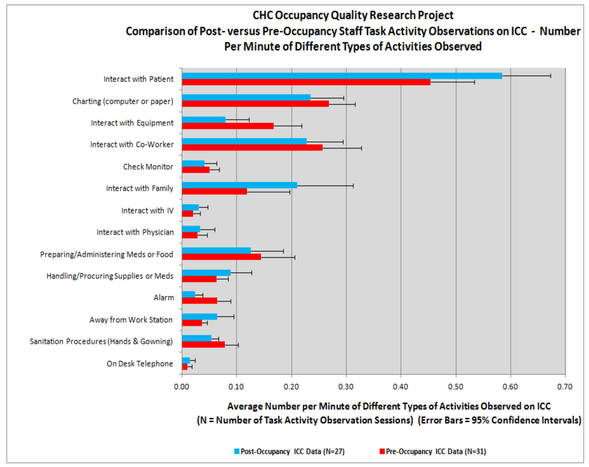


Fig. 2. Average Rankings of Major Indicators of Occupancy and Patient Care Quality by Staff in Five Different PR Survey Respondent Categories.

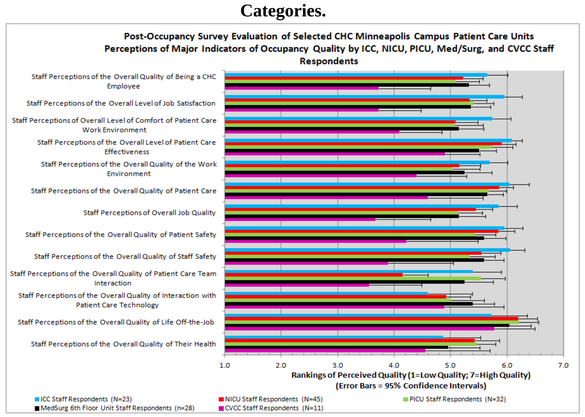


Fig. 3. Comparison of Average PR Versus MB Survey Rankings for Major Indicators of Occupancy and Patient Care Quality by ICC Respondents.

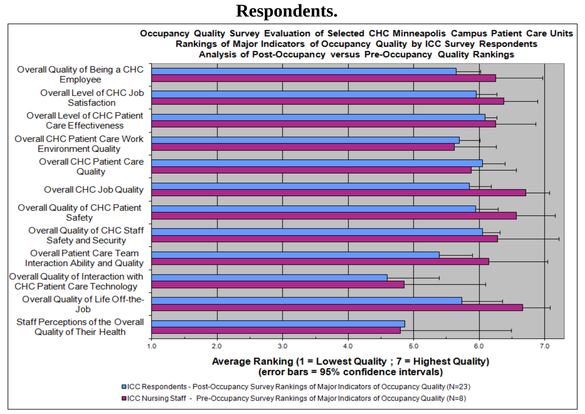


Fig. 4. Total Numbers of Concerns about Different Issues Related to the Overall Design Quality of Private Rooms, Identified in Comments by Staff Working on all Five Units.

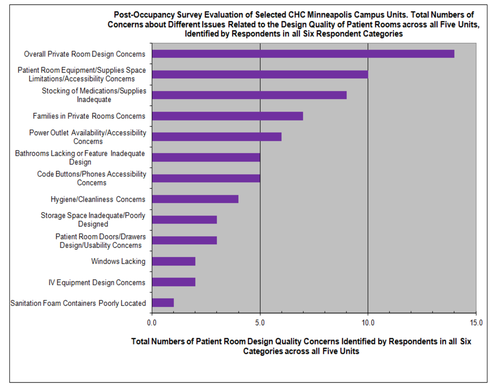
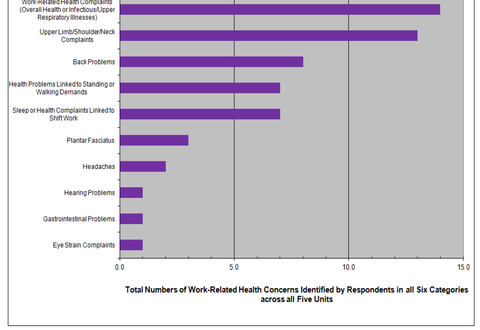


Fig. 5. Total Numbers of Concerns about Different Work-Related Health Problems, Identified in Comments by Staff Working on all Five Units.



Conclusions

► Fig. 1: Interaction with patients is significantly higher on the PR ICC relative to the MB ICC. The same pattern is observed for the NICU, PICU and Med/Surg unit. No other task activities feature significant differences between PR versus MB designs.

► Fig. 2: With 3 exceptions, PR quality rankings by CVCC staff are notably lower than PR rankings by staff on the other four units; and (2) PR rankings by staff on the other four units are roughly comparable.

► Fig. 3: All but 3 MB rankings by ICC staff , for major indicators of occupancy and patient care quality, are higher than corresponding PR rankings. A comparable pattern is observed for NICU and PICU staff (for Med/Surg staff, half of these rankings are higher for PR designs).

► Figs. 4 and 5: In recorded comments, unit staff identify a series of design shortcomings with PR patient rooms, and a series of personal health problems attributable to work in PR patient care environments.

► Do quality benefits of PR designs on the 5 different units outweigh possible quality shortcomings? Results are mixed, and suggest that CHC faces a challenge in realizing occupancy and patient care quality benefits on these PR units that match experience with NICU PR designs in other hospitals.

CEHD Research Day, March 26, 2013